PIC 10A 1A

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Today…

- Select:
  - More HW6 Hints
  - Review on Vectors
  - C++Classes
    - Concepts, Examples, and Exercise Problems

HW6 Hints

HW6 Hints (descending.cpp)

- Recall that, we wrote the array sort function that sorts an array in descending order
  - Week7_Array_Functions_Sol.cpp

- You can do the exact same thing with vectors!
- (More Hints)
  - You can compare two strings `s1` and `s2` *lexicographically*, by "<" operator
    - `s1 < s2 == true` if `s1 == "ABCDE"`, `s2 == "ABCED"`
    - So it's indeed very similar to sorting integer arrays

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HW6 Hints (mostfrequent.cpp)

- Step 1. Count the frequencies of each letter
  - Say, you have `vector<int> freq(nPossibleLetters);`

- Step 2. Find the maximum frequency
  - `int max_freq = // max frequency in freq`

- Step 3. Construct a new string, with the letters with maximum frequency

```
string answer;
for ( each character c in input ) {
    if (freq of c == max_freq) { // c has max frequency
        answer += c; // append
        // no repetition of c should be made later
        // e.g. “rgm” instead of “rgrmm” for “programming”
        // HINT: make a change s.t. [freq of c != max_freq]
    }
}
```

More hints for Step 1:
Each character has its own ASCII code, and in particular, *n*-th alphabet letter c has a property

`c-'a' == n`

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Review on Vectors

Exercise on Vectors

- Suppose that vec is a std::vector<int>, represented by
  vec == { 3, 8, -6, 0, -8, 12, 1 }.
- Determine the value of x after the code below has executed.

```
int x = vec[0];
for (auto i : vec) {
        x = ((x < i) ? i : x);
}
```

Answer: x is 12 (*max* of vec)

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Quiz #1

- Q. How do you define a vector to store salaries of 10 employees?
- Select one:
  - a. vector<double> employee_salary(10);
  - b. <double>vector employee_salary(10);
  - c. vector employee_salary(10);
  - d. double employee_salary(10);

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Quiz #1

- Q. How do you define a vector to store salaries of 10 employees?
- Select one:
  - a. vector<double> employee_salary(10);
  - b. <double>vector employee_salary(10);
  - c. vector employee_salary(10);
  - d. double employee_salary(10);

- Ans: a
  - Type of the vector is `vector<double>`
  - The size is passed to a constructor to this vector classvector<double>

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Quiz #2

- Q. What is the output of the following code snippet?
- Select one:
  - a. 0 1 2 3 4 5 6 7 8 9
  - b. 0 2 4 6 8
  - c. 1 3 5 7 9
  - d. 0 1 2 3 4

```
vector<int> myvector;
int n;
for (int i = 0; i < 10; i++)
{
    myvector.push_back(i);
    n = i % 2;
    if (n == 0)
    {
        myvector.pop_back();
    }
}
for (int i = 0; i < myvector.size(); i++)
{
    cout << myvector[i] << " ";
}
```

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Quiz #2

- Q. What is the output of the following code snippet?
- Select one:
  - a. 0 1 2 3 4 5 6 7 8 9
  - b. 0 2 4 6 8
  - c. 1 3 5 7 9
  - d. 0 1 2 3 4

```
vector<int> myvector;
int n;
for (int i = 0; i < 10; i++)
{
    myvector.push_back(i);
    n = i % 2;
    if (n == 0)
    {
        myvector.pop_back();
    }
}
for (int i = 0; i < myvector.size(); i++)
{
    cout << myvector[i] << " ";
}
```

- Ans: c
  - 1 →
  - 1 2 → 1 →
  - 1 3 →
  - 1 3 4 → 1 3 →
  - 1 3 5 →
  - 1 3 5 6 → 1 3 5 → …

*pop_back( ) called when i is even

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Quiz #3

- Q. What is the output of the following code snippet?
- Select one:
  - a. 0
  - b. 1
  - c. 2
  - d. 3

```
int s = 0;
int myarray[3];
for (int i = 0; i < 3; i++)
{
   myarray[i] = s;
   s = s + i;
}
cout << myarray[2];
```

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Quiz #3

- Q. What is the output of the following code snippet?
- Select one:
  - a. 0
  - b. 1
  - c. 2
  - d. 3

```
int s = 0;
int myarray[3];
for (int i = 0; i < 3; i++)
{
   myarray[i] = s;
   s = s + i;
}
cout << myarray[2];
```

- Ans: b
  - `myarray[0] = 0;  s = 0+0;`
  - `myarray[1] = 0;  s = 0+1;`
  - `myarray[2] = 1;  s = 1+2;`

  - Update `s` after assigning `myarray[i]`

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Quiz #4

- Q. Let print() be a member function of class Player. Complete the following code to print all elements in the vector backwards (starting from the last element and ending with the first one).

```
vector<Player> mylist;
// add elements to mylist
int i = 0;
for (___A___; ___B___; ___C___)
{
    ___D___;
}
```

- A: `i = mylist.size() - 1;`
- B: `i >= 0;`
- C: `--i;`
- D: `(mylist[i]).print();`

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Quiz #4

- Q. Let print() be a member function of class Player. Complete the following code to print all elements in the vector backwards (starting from the last element and ending with the first one).

```
vector<Player> mylist;
// add elements to mylist
int i = 0;
for (___A___; ___B___; ___C___)
{
    ___D___;
}
```

- A: `i = mylist.size() - 1;`
- B: `i >= 0;`
- C: `--i;`
- D: `(mylist[i]).print();`

- for (initialization ; condition ; update){
  // Loop body
  }

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Another Exercise

The function below has logical errors and also lacks in efficiency and robustness. Fix it on all accounts.

```
const string& concatenate_words(
  const vector<string> words ) {
    string all;
    for ( auto word : words ) {
      all += word;
    }
    return all;
}
```

Hint: one issue is similar to that of the templated max function. In this case, however, there is a universally accepted better way to manage things.

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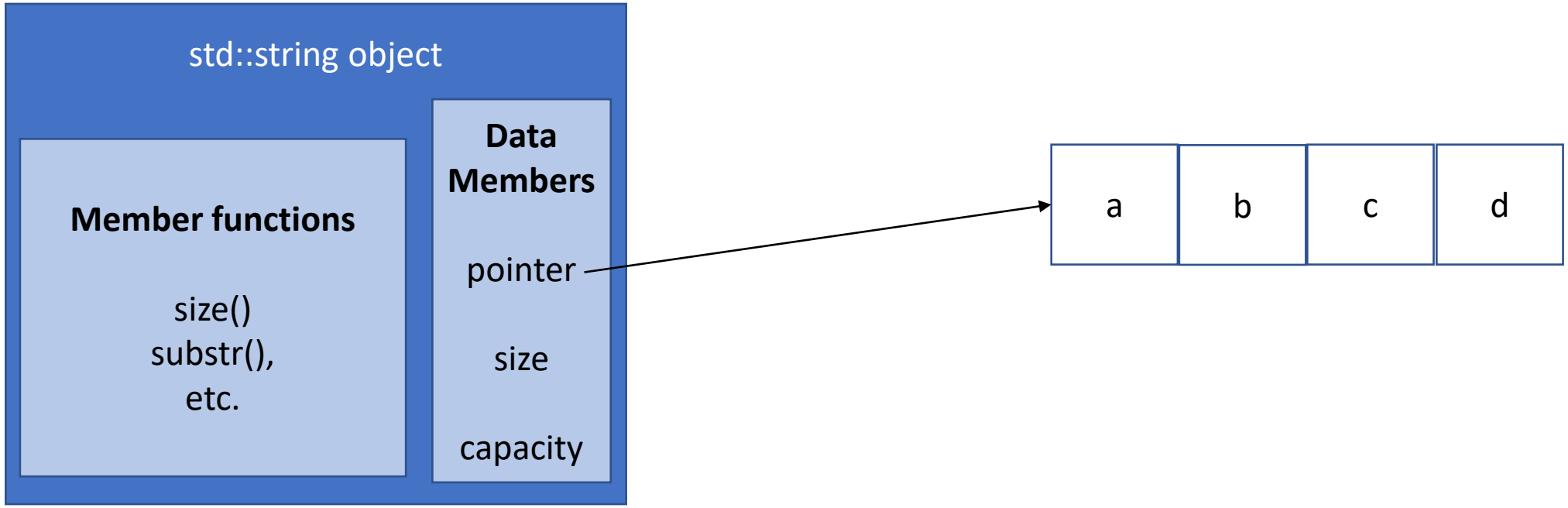
C++ Classes

Concepts, Examples, and Exercise Problems

Classes

- Roughly speaking, a class is a user-defined type that contains a collection of data members with other features (methods)
- e.g. “std::string” has data members like
  - the pointer to the char array (actual string data)
  - size and capacity variables, and some other members
  - useful member functions like size(), substr(), and operators([], ==, <, +, etc.)



Classes

- Default accessibility of data members/functions is *private* for classes
  - On the other hand, a very similar data structure, “struct” has *public* members by default

- To access the members, use the “.”(dot) operator

```
e.g.    string str = "ABCDE";
        int len = str.length();
```

- Initialization is done by a “constructor”

```
class B {
public:
    void b() const;
    B();
};
```

```
int main() {
    B b_object;
}
```

Calling the default constructor here

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Quiz #1

Consider the following code.

```
#include <iostream>
using namespace std;

class T {
 public:
  int mem;
  T() { mem = 5; }
  void fn(int x) { mem += x; }
};

int main() {
  T t1;
  t1.fn(2);
  t1.fn(3);
  cout << t1.mem << endl;
}
```

The output is

A. 0.
B. 2.
C. 3.
D. 5.
E. 10.

- Ans: E (10)

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Quiz #2

Consider the following code.

```
#include <iostream>
using namespace std;

class T {
 public:
  ...
  void fn(T t);
};

int main () {
  T t1, t2;
  ...
  t1.fn(t2); //here
}
...
void T::fn(T t) { ... }
```

The line `t1.fn(t2);` can change

A. neither `t1` nor `t2`.
B. `t1` but not `t2`.
C. `t2` but not `t1`.
D. both `t1` and `t2`.

- Ans: B
  - Note that `t2` is passed by value (copy)
  - On the other hand, the function `fn` is not marked as `const`, so it may change `t1` (its implicit parameter)

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Quiz #3

What is the output of the following code snippet?

```
class Car
{
public:
   void set_speed(double new_speed);

private:
   double speed;
};
void Car::set_speed(double new_speed)
{
   speed = new_speed;
}
int main()
{
   Car ch1;
   ch1.set_speed(144);
   cout << "Car speed: " << ch1.speed;
   return 0;
}
```

- a. Car speed: 144
- b. Car speed: 0
- c. Car speed:
- d. The code snippet does not compile.

- Ans: d
- ch1.speed is not accessible in the main() scope.

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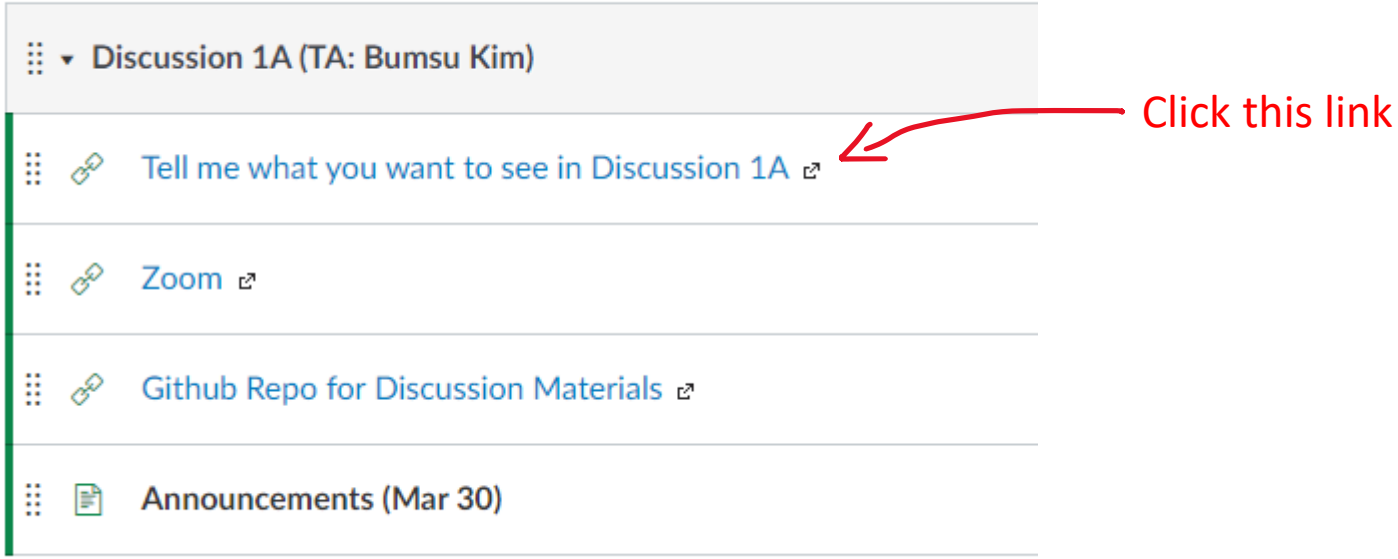
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- You can use a link in BruinLearn (Google Form)